Alex Fernsby

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Profile

Data engineer with 7 years building batch and streaming pipelines for retail and logistics. Owns shipment-event ingestion, warehouse reliability and delivery-performance reporting. Mentors two junior engineers and leads a weekly data-modelling review.

Core Competencies

- **Pipelines**: Python, SQL, Apache Airflow, Kafka and dbt for batch and streaming pipelines.
- **Cloud data platforms**: BigQuery and PostgreSQL; Docker.
- **Reliability**: Great Expectations, data contracts and operational documentation.
- **Delivery**: working knowledge of Terraform, GitHub Actions and Looker.
- **Adjacent foundations for machine learning**: tested datasets, repeatable ingestion and documented quality rules.

Professional Experience

Tidewater Logistics **Copenhagen**
Senior Data Engineer *March 2022 - present*

- Own streaming data pipelines for shipment events: 40 million events per day from Kafka into PostgreSQL and BigQuery, with Apache Airflow orchestration.
- Improved data products: cut the nightly rebuild of the delivery-performance mart from 6 hours to 55 minutes with incremental dbt models.
- Introduced data quality checks with Great Expectations on 30 core tables; failed loads dropped 70% in the first quarter.
- Moved deployments to GitHub Actions and Terraform for CI/CD; new environments take 20 minutes instead of two days.
- Defined data contracts with five producing teams and wrote the pipeline's on-call runbook for incident support.
- Cut the BigQuery events-warehouse bill by 35% through partitioning and clustering the five largest tables.
- Enabled analytics/BI consumption patterns and visualization through a Looker explore, letting planners answer delivery-delay questions themselves.
- Mentor two junior engineers and lead the weekly data-modelling review.
- Ran interviews for three engineering hires and wrote the team's take-home exercise.

Marlow Retail Group **Utrecht**
Data Engineer *June 2018 - February 2022*

- Built the company's first Airflow deployment and migrated 45 cron jobs onto it.
- Designed a PostgreSQL customer-orders star schema used by 12 analysts.
- Wrote Python ingestion services for REST and SFTP partner price feeds, handling about 2 million rows daily.
- Supported migration from on-premise PostgreSQL to BigQuery, validating 60 tables row by row.
- Trained 12 analysts on dbt and reviewed their first 40 models.
- Added alerts on 45 Airflow jobs for monitoring and observability; late feeds were noticed within 15 minutes instead of the next morning.
- Wrote the report-migration guide used by four teams moving off the old warehouse.

Delft Energy Cooperative **Delft**
Junior Analyst *September 2016 - May 2018*

- Automated monthly consumption reports with Python and SQL, saving about 3 days each month.
- Built a Flask dashboard used by 25 cooperative members to read their consumption.
- Cleaned five years of meter readings with Pandas and documented 14 data-quality rules.
- Presented consumption findings to the cooperative's board twice a year.

Education

Delft University of Technology
MSc Computer Science (data systems) *2016*

Delft University of Technology
BSc Computer Science *2014*

Projects

Open source: dbt package for freight-rate data tests; 120 GitHub stars. github.com/afernsby-example/freight-tests

2024: Talk: "Data contracts in practice", Rotterdam Data Meetup.

Certifications

2023: Google Cloud Professional Data Engineer

2022: dbt Analytics Engineering Certification

Languages

English: C1

Dutch: Native

German: A2

Publications

2024: "Testing freight data with dbt", Data Engineering Weekly guest post.

Honors and Awards

2023: Tidewater Logistics Engineering Excellence Award